Random-Slope Models

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Introduction

Random-slope models allow each cluster its own slope for one or more predictors, capturing between-cluster heterogeneity in how a covariate affects the outcome. Where random-intercept models assume that all clusters share the same effect of X but differ in baseline, random-slope models drop the shared-slope assumption and let the slope vary. They are essential for any longitudinal analysis where individual trajectories diverge over time and where the population-average slope may hide substantial subject-to-subject variability.

Prerequisites

A working understanding of random-intercept models, mixed-effects model syntax in `lme4`, and the bivariate Normal distribution for joint random effects.

Theory

For a predictor X varying within clusters:

$$Y_{ij} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i})X_{ij} + \varepsilon_{ij},$$

with $(u_{0i}, u_{1i})^\top \sim \mathcal{N}(0, \Sigma)$ where Σ contains intercept variance σ_0^2, slope variance σ_1^2, and correlation ρ_{01}. The "correlated random slope" formulation `(X | cluster)` estimates all three parameters jointly; the "uncorrelated" form `(X || cluster)` constrains $\rho_{01} = 0$.

A non-zero ρ_{01} indicates systematic dependence between baseline level and rate of change — for instance, subjects who start higher progress faster (positive correlation) or slower (negative correlation).

Assumptions

Random effects bivariate Normal, independent across clusters; conditional Normal residuals; the slope variation is real and not driven by a few influential clusters.

R Implementation

```r
library(lme4)

data(sleepstudy, package = "lme4")

# Random intercept + slope (correlated)
fit_rs <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)
summary(fit_rs)

# Uncorrelated random slope
fit_rs_unc <- lmer(Reaction ~ Days + (1 | Subject) + (0 + Days | Subject),
                   data = sleepstudy)
```

```r
# Compare with random-intercept-only
fit_ri <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)
anova(fit_ri, fit_rs)
```

Output & Results

`summary(fit_rs)` reports the population-level fixed effects, the standard deviations of the random intercept and random slope, and their correlation. `anova()` between nested models compares them via a likelihood-ratio test (with the boundary-mixture reference for variance components).

Interpretation

A reporting sentence: "Adding a random slope on `Days` substantially improved fit ($\chi^2(2) = 42.1$, $p < 0.001$); between-subject standard deviation of the slope was 5.9 ms per day, with intercept-slope correlation 0.07. The population-level slope was 10.5 ms per day (95 % CI 7.6 to 13.4)." Always report the random-slope variance and the intercept-slope correlation alongside fixed effects.

Practical Tips

- Random slopes require enough observations per cluster (rule of thumb: at least 10 per cluster); with fewer, the slope variance is unstable and may produce singular fits.
- Near-zero random-slope variance signals over-specification; drop the random slope or constrain $\rho_{01} = 0$ via the `||` syntax.
- Centring the predictor (`X - mean(X)`) often reduces intercept-slope correlation that arises mechanically from the parameterisation rather than the data.
- Check the random-effects variance-covariance with `VarCorr(fit)`; correlation estimates near ± 1 are a red flag for over-fitting.
- Use parametric bootstrap (`pbkrtest::PBmodcomp`) for small-sample LRTs on random-slope variance; the asymptotic χ^2 mixture can be unreliable.
- For non-Gaussian outcomes, the same syntax extends via `glmer()` with the appropriate family.

R Packages Used

`lme4` for `lmer()` and `glmer()`; `lmerTest` for p-values on fixed effects; `pbkrtest` for parametric-bootstrap LRTs on variance components; `glmmTMB` for an alternative implementation that handles complex error structures; `merTools` for prediction intervals that respect random-slope uncertainty.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)

- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI